

ALL-SOLID-STATE FLASH LIDAR · VSS SERIES

NO MOVING OLEI VSS-50 PARTS

An all-solid-state flash dToF sensor: a 368×152 SPAD array firing 540 000 points a second across $120^\circ \times 50^\circ$, with an IP67 all-metal housing and nothing inside it that spins.



■ WHY THIS SENSOR

- **All-solid-state flash scanning** — no rotating assembly, no bearing to wear — the reliability argument writes itself
- **$120^\circ \times 50^\circ$ field of view** — a 3D volume in front of the platform, not a single plane
- **540 000 points per second at 10 Hz** — $368H \times 152V$ SPAD array, single-echo mode
- **50 m at 90% reflectivity, 30 m at 10%** — 0.1–50 m range of measurement with ± 5 cm (1σ) accuracy
- **100k lux ambient light immunity** — direct sun, glazed façades and dark interiors on the same run
- **Built-in IMU (LSM6DS3TR-C)** — motion data alongside the cloud for mapping and odometry
- **Gigabit Ethernet, UDP output** — point cloud coordinates plus reflection intensity or reflectivity
- **IP67 all-metal housing, 720 g, DC 9–16 V** — $130 \times 75 \times 65$ mm, -20 to 55°C

0.1–50 m

RANGE

$120^\circ \times 50^\circ$

FIELD OF VIEW

540K

POINTS / SECOND

IP67

ALL-METAL HOUSING

■ APPLICATION SCENARIOS



AUTONOMOUS VEHICLES

Forward perception for on- and off-road platforms



UAV & AERIAL

Obstacle avoidance and terrain capture from the air



VOLUME & DIMENSIONING

3D capture of pallets, loads and stockpiles



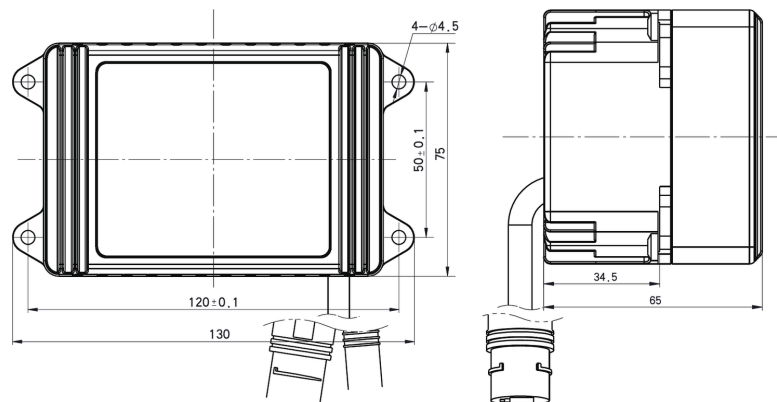
SURVEY & MAPPING

360° natural-feature and reflector mapping at range

PARAMETER	TECHNICAL DATA
Distance measurement principle	dToF distance measurement
Scanning principle	Flash scanning
Laser wavelength	905 nm
Laser safety class	Class 1 (IEC 60825-1:2014)
Measurement range	50 m @ 90% reflectivity 30 m @ 10% reflectivity
Range of measurement	0.1–50 m
Accuracy	±5 cm (1 sigma)
SPAD pixel array	368 H × 152 V
Horizontal field of view	120° (–60° to +60°)
Vertical field of view	50° (–25° to +25°)
Angular resolution	0.33° H × 0.33° V
Frame rate	10 Hz
Spot frequency (single-echo mode)	540 000 pts/s @ 10 Hz

PARAMETER	TECHNICAL DATA
Ambient light interference	100k lux
Data interface	Gigabit Ethernet
Built-in IMU	LSM6DS3TR-C
Data output protocol	UDP
UDP packet content	Point cloud coordinate values Reflection intensity or reflectivity
Power supply voltage	DC 9–16 V
Operating temperature	–20 to 55 °C
Weight	720 g
Overall dimensions	130 × 75 × 65 mm
Enclosure	IP67, all-metal housing

■ OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. Class 1 laser safety per IEC 60825-1:2014. Confirm the current EMC and regional approval status with MorpheusTEK before design-in — the manufacturer's datasheet for this product does not list EMC, vibration or shock standards.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. Range figures are stated at 90% and 10% reflectivity; the datasheet's own summary notes a maximum detection range exceeding 50 metres, while the specification table gives 0.1–50 m — quote the table. Accuracy is ±5 cm at one sigma, so plan clearances on a multiple of that, not on the figure itself. Point rate is stated in single-echo mode.

Installation. Mount the sensor so the full 120° × 50° frustum is clear of the platform's own structure; a recessed mount truncates the vertical field first. Keep the window clean — a flash sensor loses whole pixels to an obscured window rather than degrading gracefully. The housing is IP67 all-metal; allow for conducted heat at the upper end of the –20 to 55 °C range.

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